10.1 Python Pandas

Data is everywhere

We come into contact with data all the time — personally and professionally. Personally, almost all the apps on your phone rely on data — banking, social media, games, photos, email and more. We also see data in commerce and science, and, with the decreasing cost of accessing it, the quantity of data is growing extremely quickly. Put simply, data science helps us deal with data; more formally, data science is a combination of multiple disciplines that use statistics, data analysis and machine learning to analyse data and extract knowledge and insights from it. In this module, we learn some tools and techniques that have been built for Python to help work with data.

Watch: Pandas to manipulate data [23.29 mins]



[Transcript (https://learn.usc.edu.au/courses/35243/files/3137620?wrap=1)](https://learn.usc.edu.au/courses/35243/files/3137620?wrap=1)

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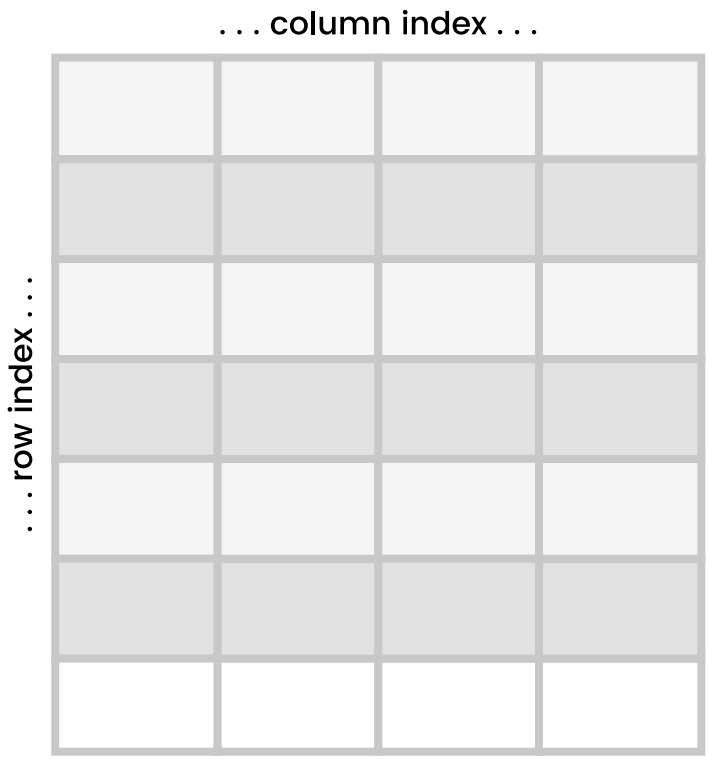
What is a Pandas DataFrame?

A DataFrame object is a 2D table of values (see figure 10.1).

Like a spreadsheet, a DataFram has:

- an *ordered* collection of columns
  - each column can have a different type
  - values within the same column should have the same type
- an index for the rows (just like Series)
- an index for the columns (column labels).

Figure 10.1 Pandas DataFrame



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Advantages of Pandas DataFrames include the following:

- They have an optional *row-index* and *column-index*.
- They are very easy to read from a CSV file.
- They include lots of powerful query and reduce functions.
- New columns are easy to calculate.

- Hundreds/thousands of Excel formulae can be added in a single line of Python code!
- Indexes are automatically aligned when you combine DataFrames and Series.
- They integrate nicely with Matplotlib to get useful visualisations.